

MATH 118 HOMEWORK ASSIGNMENT 5

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Section 2

Q1) 1.a) $\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$ $U_{y|x} = Bx$

x_i	y_i	x_i^2	$x_i y_i$
0.5	1.3	0.25	0.65
1.5	3.4	2.25	5.10
3.2	6.7	10.24	21.64
4.2	8.0	17.64	33.60
5.1	10.0	26.01	51.00
6.5	13.2	42.25	85.80
Sum(Σ)		98.64	197.59

$$\hat{\beta} = \frac{197.59}{98.64} \approx 2.0031$$

so; $\hat{y} = 2.0031x$

1.b) The model is $U_{y|x} = \beta_0 + \beta_1 x$ so we will use;

$$\bar{x} = \frac{\sum x_i}{n} \quad \bar{y} = \frac{\sum y_i}{n}$$

$$s_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n} \quad s_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}$$

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$n=6$$

$$\sum x_i = 0.5 + 1.5 + 3.2 + 4.2 + 5.1 + 6.5 = 21 \Rightarrow \bar{x} = \frac{21}{6} = 3.5$$

$$\sum y_i = 1.3 + 3.4 + 6.7 + 8.0 + 10.0 + 13.2 = 42.6 \Rightarrow \bar{y} = \frac{42.6}{6} = 7.1$$

$$\sum x_i^2 = 98.64$$

$$\sum x_i y_i = 197.59$$

$$s_{xx} = 98.64 - \frac{(21)^2}{6} = 98.64 - 73.5 = 25.14 = s_{xx}$$

$$s_{xy} = 197.59 - \frac{(21)(42.6)}{6} = 197.59 - 149.1 = 48.49 = s_{xy}$$

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} = \frac{48.49}{25.14} \approx 1.9288$$

$$\hat{\beta}_0 = 7.1 - (1.9288)(3.5) = 7.1 - 6.7508 = 0.3492 = \hat{\beta}_0$$

$$\hat{y} = 0.3492 + 1.9288x$$

1.b) Hypothesis Test

$$H_0: \beta_0 = 0$$

$$H_1: \beta_0 \neq 0$$

$$\alpha = 0.10$$

SE (standard error of $\hat{\beta}_0$):

$$S_{yy} = \sum y^2 - n\bar{y}^2 = 396.38 - 6(7.1)^2 = 93.92$$

$$SSE = S_{yy} - \beta_1 S_{xy} = 93.92 - (1.9288)(48.49) \approx 0.3925$$

$$MSE = \frac{SSE}{n-2} = \frac{0.3925}{4} \approx 0.0981$$

$$SE(\hat{\beta}_0) = \sqrt{MSE \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} = \sqrt{0.0981 \cdot \left(\frac{1}{6} + \frac{(3.5)^2}{25.14} \right)} \approx 0.2532$$

$$t = \frac{\beta_0 - 0}{SE(\hat{\beta}_0)} = \frac{0.3492}{0.2532} \approx \boxed{1.379}$$

Decision

1) Critical value

Significance level (α) = 0.10

$$df = n - 2 = 4$$

Test type = Two tailed (since $H_1: \beta_0 \neq 0$)

$$\alpha/2 = 0.05 \text{ with } df = 4$$

$$t_{crit} = t_{0.05, 4} = 2.132$$

calculated t -value ≈ 1.379

critical t -value = 2.132

Comparison Rule: since $|t_{calc}| < t_{crit}$ ($1.379 < 2.132$) the calculated value does not fall into the rejection region

• Fail to reject the null hypothesis (H_0)

Conclusion: At the 0.10 level of significance there is insufficient evidence to conclude that intercept β_0 is different from zero.

This suggests that the model passing through the origin ($\beta_0 = 0$) is a more appropriate representation of the data.

Q2) 2a) $n=10$, $\sum x = -2678.6$, $\sum y = 2.007$, $\sum x^2 = 717826.16$

$\sum xy = -1892$

$\bar{x} = \frac{-2678.6}{10} = -267.86$

$\bar{y} = 0.2007$

$S_{xx} = 336.364$

$S_{xy} = -15.1114$

Coefficients

slope $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-15.1114}{336.364} \approx -0.0449$

$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.2007 - (-0.0449)(-267.86)$

$\hat{\beta}_0 = -11.326$

$\hat{y} = -11.326 - 0.0449x$

2b) Yes the data show that the proportion of impurities increases at the temperature approaches -273°C . The fitted linear regression has a negative slope, indicating that lower temperatures are associated with higher impurity proportions. This supports the conclusion that impurities increase as the temperature approaches -273°C .

2c)

$R^2 = \frac{S_{xy}^2}{S_{xx} \cdot S_{yy}}$

$n=10$, $\sum x = -2678.6$, $\sum y = 2.007$

$\cdot S_{xx} = 336.364$

$\cdot S_{yy} = 0.7195$

$\cdot S_{xy} = -15.1114$

$R^2 = \frac{(-15.1114)^2}{(336.364)(0.7195)} \Rightarrow R^2 \approx 0.9436$

2d) Yes, the linear model appears highly appropriate for 2 main reasons;

- model explains 94% of data variation
- clear trend

To fully validate the model we would need:

- Residual Plot
- Significance testing, Physical Limits

Q3) 3a) $n=10$

Variable	Total (Σ)	Mean ($\bar{x}$)
y	825	82.5
x_1	379	37.9
x_2	2417	241.7

$$\Sigma x_1^2 = 14533$$

$$\Sigma x_2^2 = 601343$$

$$\Sigma x_1 y = 31562$$

$$\Sigma x_2 y = 202758$$

$$\Sigma x_1 x_2 = 92604$$

$$1) n\beta_0 + \beta_1 \Sigma x_1 + \beta_2 \Sigma x_2 = \Sigma y$$

$$2) \beta_0 \Sigma x_1 + \beta_1 \Sigma x_1^2 + \beta_2 \Sigma x_1 x_2 = \Sigma x_1 y$$

$$3) \beta_0 \Sigma x_2 + \beta_1 \Sigma x_1 x_2 + \beta_2 \Sigma x_2^2 = \Sigma x_2 y$$

From the equations

$$\hat{\beta}_0 \approx -0.835, \hat{\beta}_1 \approx 1.144, \hat{\beta}_2 \approx 0.165$$

$$\hat{y} = -0.835 + 1.144x_1 + 0.165x_2$$

3b)

Substitution of values:

$$x_1 = 38$$

$$x_2 = 235$$

$$\hat{y} = -0.835 + 1.144(38) + 0.165(235)$$

$$\hat{y} = -0.835 + 43.472 + 38.775$$

$$\hat{y} = 81.412 \text{ kg}$$

The predicted final weight for an animal with an initial weight of 38 kg and feed consumption of 235 kg is approximately 81.41 kg